

Field Troubleshooting Template Pack

Companion downloads for **Instrumentation Troubleshooting: Loops, Transmitters & Valves**

Print single-sided. Each form fits one page.

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Safety notice: These forms do not authorize work. Follow your site's permit, lockout/tagout, energized-work, and management-of-change procedures. When a form step conflicts with site procedure, site procedure wins.

1. Five-Question Triage Card

Fill this out **before** you pick up a tool.

Tag: _____ **Date/Time:** _____ **Tech:** _____

Q1. What changed? Last known good: _____ Work/MOC/weather/startup since: _____

Q2. What does the operator see? Reported symptom (their words): _____ Process actually doing: _____

Q3. What does the controller see? PV: _____ Raw counts: _____ Scaling: _____ to _____ Alarms / first-out: _____ Mode: AUTO / MAN / CAS

Q4. What does the meter see? Loop current: _____ mA Loop voltage: _____ V Supply: _____ V

Q5. What does the device see? Actual process condition at sensor: _____ Device local display / HART PV: _____

Disagreement check — circle where the chain breaks

Process → Sensor → Transmitter → Wiring → Card → Controller → Display

Mark the link where reality and indication stop agreeing. Everything upstream of that mark is proven good.

First hypothesis: _____ **Measurement that will prove or kill it:**

2. Split-Half Isolation Worksheet

The loop is a chain. Cut it in the middle, not at the ends.

Tag: _____ **Date:** _____

Step	Measure at	Expected	Actual	Good / Bad	Halves remaining
1	Controller input (raw counts)				
2	Field terminal / marshalling				
3	Junction box				
4	Transmitter terminals				
5	Sensor / process connection				

Rule: after each measurement, you should be able to eliminate at least half the remaining chain. If a measurement eliminates nothing, it was the wrong measurement.

Fault isolated to: _____ **Evidence (the two readings that bracket it):**
_____ **Verification after repair:** _____

3. 4–20 mA Reference & Conversion Sheet

Signal-to-percent table

mA	4.0	5.6	7.2	8.0	8.8	10.4	12.0
% span	0	10	20	25	30	40	50

mA	13.6	14.0	15.2	16.0	16.8	18.4	20.0
% span	60	62.5	70	75	80	90	100

Math both directions

- Percent from current: $\% = (I - 4) / 16 \times 100$
- Current from percent: $I = 4 + 0.16 \times \%$
- Engineering units: $EU = LRV + ((I - 4) / 16) \times (URV - LRV)$
- Current from EU: $I = 4 + 16 \times (EU - LRV) / (URV - LRV)$

What an out-of-band reading tells you

Reading	Most likely meaning	Next check
0.0 mA	Open circuit, blown fuse, dead supply, lifted wire	Loop voltage at both ends
Above 0 but at or below 3.6 mA	Device failure alarm (NAMUR NE43 low), or supply/leakage problem	HART status, supply at terminals
3.6–3.8 mA	Ambiguous zone — not valid measurement, not a clean alarm	HART status, configured alarm levels
3.8–4.0 mA	Under-range saturation — process below LRV or range wrong	Actual process value, LRV setting
4.0 mA exactly, never moves	Fixed-current/loop-test mode, wrong range, or process truly at LRV	HART mode, independent indication
20.0–20.5 mA	Over-range saturation — process above URV or range wrong	Actual process value, URV setting
21.0 mA or higher	Device failure alarm (NAMUR NE43 high)	HART diagnostics, sensor connection
Well above ~23 mA	Beyond normal transmitter output — suspect short or foreign source	Isolate and meter transmitter alone
Correct at device, wrong at controller	Wiring, card, or scaling	Raw counts and scaling config

Alarm levels are configurable and vendor defaults differ (some ship at 3.75 / 21.75 mA). Read the device's configured alarm direction and levels before trusting this table.

Burden and supply quick checks

- Supply voltage at transmitter terminals must exceed device minimum (commonly 10.5–12 V DC) **at 20 mA**, not just at 4 mA.
- Sense resistor: 250 Ω gives 1–5 V across 4–20 mA.
- Total loop resistance must stay below the supply's stated maximum burden.

4. Loop Check Form (Cold and Hot)

Loop tag: _____ **Service:** _____ **Date:** _____ **Range:** _____ to _____ (units: _____)

Fail direction: UP / DOWN / HOLD

Pre-check

Item	Verified	Initials
Permit / LOTO in place as required	☐	
Operations notified, loop out of service or bypassed as approved	☐	
Tag, terminal numbers, and drawing revision match the field	☐	
Wiring terminations tight, shield grounded at one end only	☐	
Supply voltage at transmitter: _____ V DC	☐	

Cold loop check (continuity / wiring, de-energized)

From	To	Expected	Actual	Pass
				☐
				☐
				☐

Hot loop check (signal injection, five points)

Point	Injected	Expected mA	Actual mA	Controller reads	Expected EU	Pass
0%		4.00				☐
25%		8.00				☐
50%		12.00				☐
75%		16.00				☐
100%		20.00				☐

Alarm and interlock verification

Function	Setpoint	Tested value	Acted correctly	Pass
Low alarm			☐	☐
High alarm			☐	☐
Trip / interlock			☐	☐

Restore

Item	Done
Bypasses removed, alarms re-enabled	☐
Loop returned to normal mode, operations confirmed	☐
Documentation filed, CMMS closed	☐

Technician: _____ **Verified by:** _____

5. As-Found / As-Left Calibration Record

Tag: _____ Device: _____ S/N: _____ Type: Pressure / DP / Level / Flow / Temp / Other: _____
Range (LRV-URV): _____ to _____ Units: _____ Tolerance: $\pm$ _____ % of span
Interval: _____ Test standard used: _____ Std. cal. due: _____ Date: _____ Technician: _____
Ambient: _____ °F / °C

As-found (do not adjust before recording)

% Span	Input applied	Ideal output	As-found output	Error (% span)	In tol.
0					☐
25					☐
50					☐
75					☐
100					☐
50 (down)					☐
0 (down)					☐

Max as-found error: _____ % span Hysteresis observed: _____ % span

Action taken

☐ No adjustment needed ☐ Zero trim ☐ Span trim ☐ Sensor trim ☐ Re-range only ☐ Repaired: _____ ☐
Replaced: _____

As-left

% Span	Input applied	Ideal output	As-left output	Error (% span)	In tol.
0					☐
25					☐
50					☐
75					☐
100					☐

Max as-left error: _____ % span Result: PASS / FAIL

Notes and follow-up

Technician sign-off: _____ Reviewed: _____

6. Case Study Worksheet

Use this for the filmed faults and for real calls. One fault per sheet.

Tag / equipment: _____ **Date:** _____

1. Symptom as reported _____

2. Symptom as observed _____

3. Hypotheses (list at least three before measuring)

	#	Hypothesis	Measurement that would disprove it
	1		
	2		
	3		

4. Measurements taken (in order)

	#	What / where	Result	Hypotheses eliminated
	1			
	2			
	3			
	4			

5. Root cause and the evidence that proves it _____

6. Repair performed _____

7. Verification — how you proved it's fixed _____

8. What would have found this faster? _____

7. Master Diagnostic Checklists by Symptom

Reading pegged low (at or below 4 mA)

1. **Meter the loop current** at the field terminals, not the controller.
2. **Check loop voltage** at the transmitter with the loop closed.
3. **Confirm polarity** and that no wire is lifted or backed out.
4. **Read the device locally** (display or HART) — does it agree with the mA?
5. **Compare to the actual process** with an independent indication.
6. **Check for a low failure alarm** (at or below 3.6 mA under NAMUR NE43, or the device's configured low alarm = device declaring its own fault).
7. **Verify range and units** haven't been changed.
8. *Failure mode:* declaring a dead transmitter when the supply fuse is open.

Reading pegged high (at or above 20 mA)

1. Meter loop current; distinguish 20.0 (saturated) from 21+ (fault/short).
2. Check for a short across the loop or a pinched cable.
3. Read HART diagnostics for upscale burnout and sensor limits.
4. Verify the process isn't genuinely over-range.
5. For DP level, check for a lost wet leg or plugged low-side tap.
6. For temperature, check for an open thermocouple or RTD lead.
7. *Failure mode:* re-ranging to hide a real over-range condition.

Reading is wrong but stable

1. Compare device local reading, controller PV, and independent measurement — the disagreement locates the fault.
2. Check controller **scaling and raw counts** before touching the field device.
3. Check for zero shift: has the device been overpressured, or the mounting changed?
4. Verify installation: impulse line slope, sealing fluid, thermowell contact, probe fouling.
5. Perform as-found calibration before adjusting anything.
6. *Failure mode:* calibrating a good transmitter because the scaling was wrong.

Reading is noisy or jumping

1. Look at raw counts, not the filtered PV — is the noise real?
2. Check shield grounding: one end only, no double ground.
3. Check for VFD or motor cabling shared in the same tray or conduit.
4. Check terminations for corrosion and loose screws; wiggle-test while watching.
5. Check for genuine process turbulence, flashing, or agitation.
6. Only then increase damping — damping hides faults.
7. *Failure mode:* adding damping until the symptom disappears and the fault stays.

Reading is slow or sticky

1. Check filtering/damping settings in device and controller.
2. Check for plugged impulse lines, frozen sensing lines, or fouled probes.
3. For temperature, check thermowell fit, air gap, and paste.
4. Confirm scan rate and update time in the controller.

Valve won't move or won't reach setpoint

1. **Verify the signal arriving at the positioner** in mA, at the positioner.
2. **Check air supply pressure** at the regulator, under load, while stroking.
3. Confirm fail direction and that no mechanical stop or handwheel is engaged.
4. Stroke locally at the positioner to separate signal problems from mechanical.
5. Check positioner calibration (zero and span of travel).
6. Check solenoid: energized state, coil continuity, spool free.
7. Check packing friction and stem alignment.
8. *Failure mode*: recalibrating a positioner when the air regulator sags under flow.

Valve hunts or cycles

1. Determine whether the demand signal itself is oscillating — if yes, it's upstream.
2. Check for excessive positioner gain.
3. Check packing friction and backlash (deadband on reversal).
4. Check air supply capacity and booster/volume-tank sizing.
5. Only then look at PID tuning.

Intermittent trip or alarm

1. Get the **first-out / time-stamped event** before theorizing.
2. Correlate with process events, startups, ambient temperature, and rain.
3. Wiggle-test terminations and flex points while monitoring.
4. Check switch setpoint versus normal operating margin — is it simply too close?
5. Check for moisture in junction boxes and conduit seals.
6. Install a recorder or trend the raw signal at high resolution and wait.
7. *Failure mode*: replacing the switch, seeing no recurrence for a week, and calling it fixed.